

Projexify

AI-Powered GitHub Project Recommendation System

Developed by **Nidhi Kumari** ·

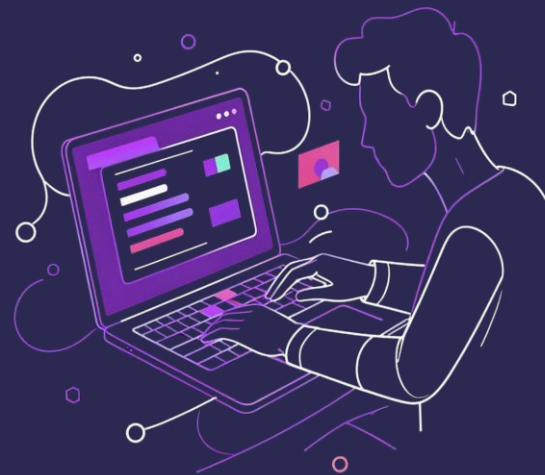
BTech, Artificial Intelligence & Machine Learning

Noida Institute Of Engineering And Technology.

AI/ML PROJECT

GITHUB INTEGRATION

REACT + NODE.JS



The Problem — and the Solution

The Challenge Students Face

GitHub hosts over 300 million repositories — a treasure trove of learning opportunities, but also an overwhelming maze for students and junior developers. Finding a project that matches your exact skill set, complexity preference, and learning goals is like finding a needle in a haystack.

- Students struggle to identify projects that align with their current skill level
- Manual searching is time-consuming and often leads to irrelevant results
- No existing tool provides truly personalized, AI-driven project recommendations
- Beginners frequently end up with projects that are either too simple or too complex

Projexify — The Smart Solution

Projexify bridges the gap between a student's skill set and the vast GitHub ecosystem using the power of artificial intelligence. Rather than browsing aimlessly, users simply input their skills and let the AI do the heavy lifting — surfacing the most relevant, real-world repositories in seconds.

- AI-driven recommendations tailored to your specific skill profile
- Pulls real-time repositories directly from the GitHub API
- Each suggestion comes with a concise description and a direct GitHub link
- Designed to accelerate learning and reduce discovery friction

Objectives

Projexify was built around a clear set of goals — each one designed to make project discovery faster, smarter, and more meaningful for students and developers at every stage of their journey.

1

Skill-Based Input

Accept the user's current technical skills as input, forming the foundation for all downstream recommendations. The system is designed to handle a wide range of technologies and domains.

2

Real-Time GitHub Fetch

Integrate directly with the GitHub API to retrieve live, up-to-date repositories — ensuring users always see active and maintained projects rather than stale or archived ones.

3

AI-Powered Filtering

Apply large language model intelligence via the Groq API to intelligently analyze, rank, and filter repositories so only the most relevant results reach the user.

4

Actionable Suggestions

Present each recommended project with a short, human-readable description and a direct GitHub link, enabling users to evaluate and jump into a project immediately.

5

Enhance Discovery

Transform the project discovery experience from a tedious manual search into a streamlined, intelligent workflow — saving hours and improving learning outcomes for students.

Technologies Used

Projexify is built on a modern, full-stack architecture — combining a responsive React frontend, a robust Node.js backend, and cutting-edge AI APIs to deliver seamless, intelligent recommendations.



Frontend

- **React JS** — Component-based UI for dynamic, responsive views
- **CSS** — Custom styling for a clean, modern look and feel
- **Axios** — Promise-based HTTP client for seamless API communication
- **React Icons** — Rich icon library for an intuitive interface



Backend

- **Node.js** — Lightweight, event-driven JavaScript runtime for the server layer
- **Express.js** — Minimal and flexible web framework for building RESTful API routes

The backend acts as the orchestration layer — receiving user inputs, querying the GitHub API, passing data to the AI model, and returning curated results to the frontend.



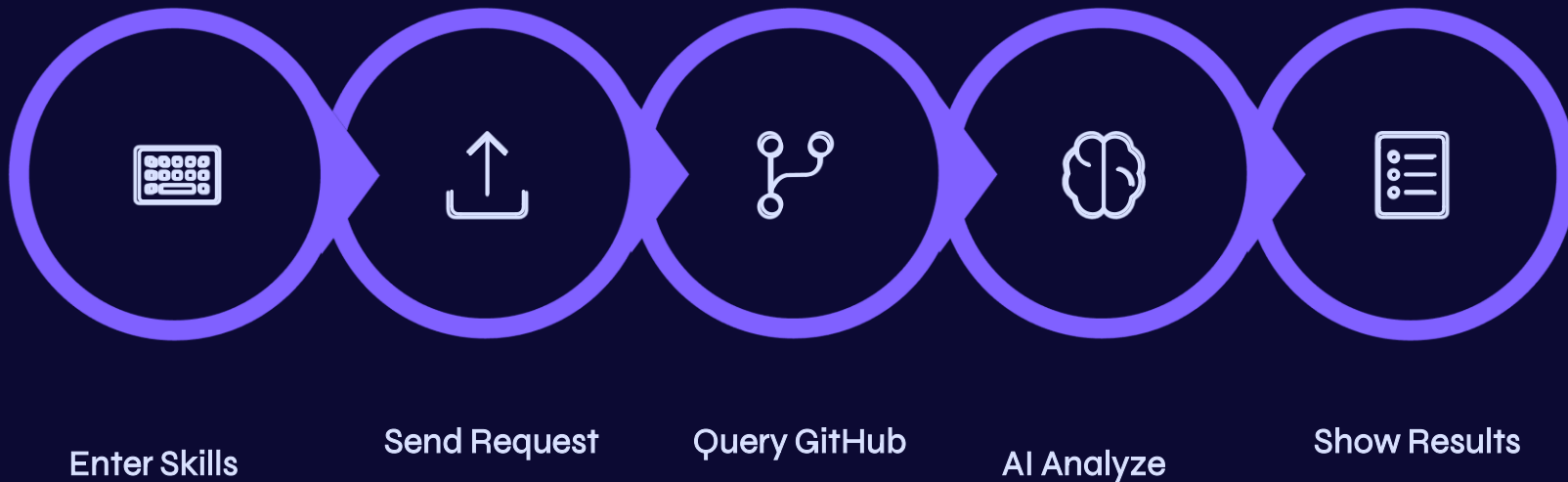
APIs & AI

- **GitHub API** — Real-time access to millions of public repositories with rich metadata
- **Groq API (Llama Model)** — High-speed inference using Meta's Llama large language model for intelligent repository analysis and ranking

The Groq API enables near-instantaneous LLM responses, making the recommendation experience fast and fluid.

System Workflow

Projexify follows a clean, end-to-end pipeline — from the moment a user enters their skills to the moment polished recommendations appear on screen. Every step is optimized for speed, accuracy, and relevance.



The entire pipeline is designed to be fast and transparent. The frontend communicates with the backend via Axios HTTP calls, the backend orchestrates GitHub data fetching and AI inference in parallel where possible, and the final output is rendered in an interactive card-based UI — giving users everything they need to evaluate and explore a project at a glance.

Features of Projexify



Skill-Based Recommendations

Tailored suggestions based on the exact technologies and languages the user specifies — no noise, no irrelevant results.



AI-Powered Filtering

Leverages the Groq Llama model to intelligently analyze repository content, README quality, and relevance before surfacing results.



Real GitHub Integration

All recommended projects are live, real repositories pulled directly from GitHub — not curated lists or static datasets.



Description + Direct Link

Each recommendation includes a short AI-generated summary and a one-click GitHub link so users can dive in immediately.



Responsive Interface

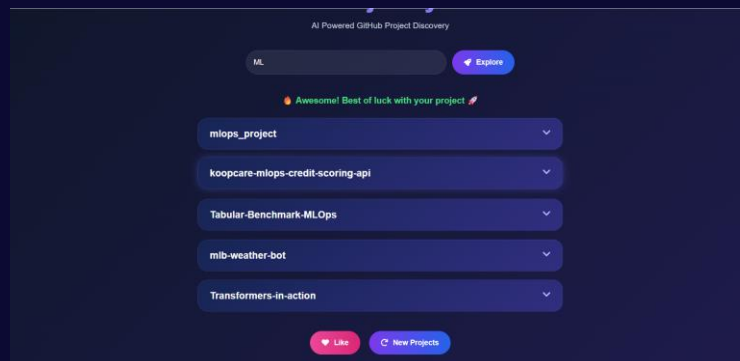
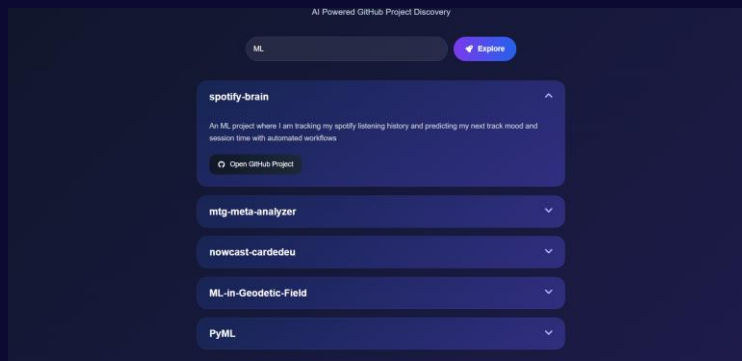
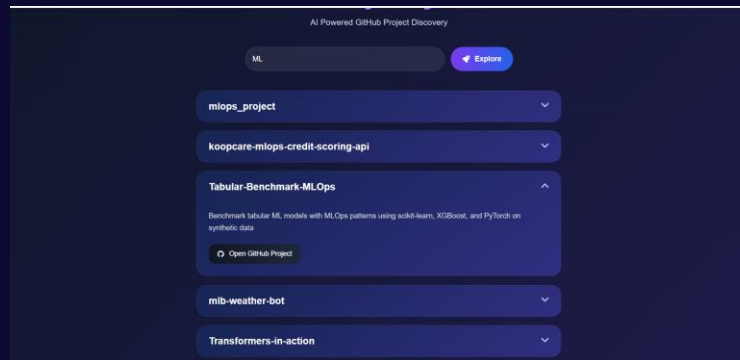
A clean, responsive UI that works seamlessly across desktop and mobile devices, making discovery accessible anywhere.



Fast Discovery

The combination of Groq's high-speed inference and efficient API calls means users get curated results in seconds, not minutes.

Live Demonstration



Future Scope

Projexify lays a strong foundation, but the roadmap ahead is rich with opportunities to grow into a full-featured learning and discovery platform. Here are the key directions planned for future development.

User Profiles & Authentication

Add login and profile management so users can save their skill sets, revisit past recommendations, and track their learning journey over time.

Enhanced ML Recommendation Models

Replace or augment the current LLM-based filtering with fine-tuned ML models trained on user behavior and feedback data to improve long-term recommendation accuracy.

Multi-Platform Expansion

Extend recommendations beyond GitHub to include platforms like **Kaggle** (for data science projects) and **Hugging Face** (for AI/ML model repositories), broadening the scope of discovery.

Bookmark & Save Feature

Enable users to bookmark interesting projects and organize them into collections, making it easy to return to shortlisted repositories later.

Rating & Feedback System

Allow users to rate recommended projects and provide feedback, creating a feedback loop that continuously improves the AI model's relevance scoring.

Skill-Based Learning Roadmaps

Go beyond project suggestions by generating personalized learning roadmaps — showing users which skills to acquire next and which projects to tackle in sequence.

Cloud Deployment

Deploy Projexify as a fully hosted web application on cloud infrastructure (AWS, Vercel, or Railway) with a public URL, making it accessible to students worldwide.

Resume-Based Recommendations

Allow users to upload their resume or LinkedIn profile, enabling the AI to automatically extract skills and generate targeted project recommendations without manual input.

Conclusion

Projexify represents a meaningful step forward in how students and junior developers discover, evaluate, and engage with open-source projects on GitHub. By combining real-time data with AI intelligence, it turns an overwhelming search problem into a streamlined, personalized experience.

Solves a Real Problem

Directly addresses the pain point of information overload on GitHub, giving students a targeted, skill-aligned starting point for their learning journey.

AI-Driven Accuracy

The integration of Groq's Llama model elevates the system beyond simple keyword search, delivering contextually relevant recommendations that reflect actual project suitability.

Saves Time

What might take hours of manual browsing is compressed into seconds — freeing up students to focus on building and learning rather than searching.

Scalable Foundation

The modular, API-first architecture means Projexify can easily scale — adding new platforms, richer ML models, user profiles, and cloud deployment with minimal architectural rework.

Projexify makes project discovery simple, intelligent, and genuinely useful — empowering every student to find the right project at the right time and accelerate their growth as a developer.